

Teaching Tech Together

Teaching Online



Greg Wilson
<https://third-bit.com>



Estimated time: 40 minutes.

All art from openclipart.org (open license)

Learning Objectives

1. Describe and summarize techniques for making interactive online teaching more effective.
2. Describe two methods for ensuring that attention is fairly distributed among class participants and explain why this is important.
3. Describe at least four different kinds of exercises that students can do in a programming class beyond simply writing and running code.
4. Describe four ways to accommodate learners of widely-varying backgrounds in a single class and explain the pros and cons of each.

Don't assume—ask

Don't presume people have {thing}, have great {thing}, have {thing} to themselves, or even have a home where {thing} could be. For each thing, fill in: internet, computer, monitor, attention, etc.
— Elizabeth Wickes

Don't make assumptions about your learners' technology or circumstances. As Elizabeth Wickes said, "Don't presume people have {thing}, have great {thing}, have {thing} to themselves, or even have a home where {thing} could be. For each thing, fill in: internet, computer, monitor, attention, etc." And remember their schedule has undoubtedly changed on short notice.

<https://elizabethwickes.com/2020/03/12/tips-for-live-teaching-tech-online-deeply-informed-by-the-carpentries/>

Don't change what you don't have to



Long walk?
Old shoes.

Don't change what you don't have to. Assignments, submission, grading, discussion forums: if you're already doing them online, don't change them now. The same goes for content: your slides and exercises may not be perfect, but nothing ever is.

<https://unsplash.com/photos/x7gz40Z9ObM>

Don't try to be (or beat) recorded video

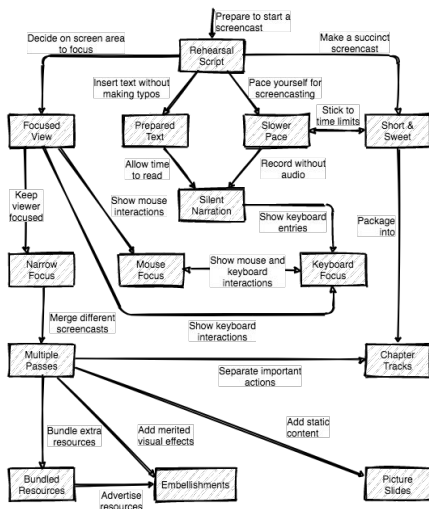


Requires
time and
money

Don't try to be (or to beat) recorded video. Even mediocre videos require time and production experience that you probably don't have right now, so move your live teaching online instead of trying to record and edit yourself.

<https://unsplash.com/photos/t4ql2lDcL5s>

But if you do...



Chen & Rabb:
“A Pattern Language for Screencasting.”
<https://doi.org/10.1145/1943226.1943234>

If you are going to make videos, please read Chen and Rabb's "A Pattern Language for Screencasting" first.

<https://doi.org/10.1145/1943226.1943234>

Learn the song, then improvise over it



← The mistakes
are the
pedagogy

6

Learn the song, then improvise over it. Good musicians don't just make things up on the fly: they learn the song and then improvise around it. Great musicians listen to their bandmates and their audience while they do this. The best live teaching (whether in person or online) works the same way: make a plan, then vary your delivery depending on who your learners are, what questions they ask, and what you realize as you're speaking.

<https://unsplash.com/photos/V3u2hyLPbaM>

Don't teach alone



Every great surgeon has a great nurse beside her

7

Don't teach alone. Every great surgeon has a great nurse beside her. Equally, when you're teaching online for the first time, get someone else to monitor the chat for questions, figure out who to call on next, and so on. If none of your colleagues can do this (remember, they're probably scrambling too), ask one of your more advanced learners to help out. Whoever it is, make sure to introduce them at the start of class.

<https://unsplash.com/photos/qLzWvcQq-V8>

Do a dry run (with your copilot)



*"Nobody told
me there was
a firewall..."*

Do a dry run (with your copilot). Fifteen minutes with a small group before you do a real (larger) class can save you from saying, "Nobody told me there was a firewall."

Direct traffic



←
Maybe not
this much...

Direct traffic. Tell people where to put things and where to direct their attention. "Put it in the chat," "add it to the doc," and "unmute and tell us" will all make the class run much more smoothly.

<https://www.youtube.com/watch?v=MXuDMKCHnDc>

Don't teach to empty chairs



← This is hard

10

Don't teach to empty chairs. We all rely on small social cues like nods and "mm hms" far more than we realize. Teaching without these is really hard, so if it's not practical to have everyone turn on video, choose a subset of 4-6 learners and have them do it so that your hindbrain believes you actually have an audience. This also promotes engagement: learners can see that they're not alone, and after each break, you can select a different subset of learners and direct your questions to them.

<https://unsplash.com/photos/3XSyPucMF3c>

Mute early, mute often



Have people
raise hands
to speak

11

Mute early, mute often. Background noise kills meetings, so mute everyone after every speaker and then unmute the next person you've called on. And as [Chelsea Troy pointed out](#), in most online meetings, the first person to speak during a pause gets the floor. The result? "If you have something you want to say, you have to stop listening to the person currently speaking and instead focus on when they're gonna pause or finish so you can leap into that nanosecond of silence and be the first to utter something. The format...encourages participants who want to contribute to say more and listen less." The solution is to run a text chat beside the video conference where people signal that they want to speak, and then call on them in turn.

<https://unsplash.com/photos/BcjdbyKWquw>

Call on people by name



Most people
won't do
this

12

Call on people by name. "Any questions?" often produces an awkward silence, and even when it doesn't, it gives fluent extroverts undue air time. Instead, call on people from your selected group by name. (If your video conferencing tool doesn't display real names by default, ask participants to give theirs when they ask questions in the chat.) This also helps even out the effects of differing ability: you can throw easy questions at people who are struggling and save the hard ones for people who might otherwise get bored.

<https://unsplash.com/photos/AoqgGAqrLpU>

Use breakout rooms



Helps keep
learners
engaged

13

<https://unsplash.com/photos/1K8plblrhkQ>

Use breakout rooms. Peer instruction is the most effective scalable teaching practice we know of. It relies on students debating their answers to a motivating question for two or three minutes at a time. Zoom's breakout rooms make this relatively easy to run online: it takes 15-30 seconds to get everyone into rooms, and in a class of forty there will be one or two who initially have problems, but it does a lot to keep everyone engaged and motivated. (Zoom also supports whiteboards, but please don't try to use them until you're comfortable with its other features.)

Add speaker's notes to your slides



←
School is
difficult
enough

Add speaker's notes to your slides. These can be point-form or long-form depending on how much time you have, but they will make your content easier to find, easier to review, and more accessible to people who have trouble seeing or hearing.

Use accessibility tools to help everyone



Helps
everyone

Use accessibility tools to help everyone. Speaker's notes are just one example of this: enlarging your cursor, using a [key caster](#) to display control keys as you type, switching to high-contrast mode all help compensate for shaky video or low-quality screens.

Invest in a good microphone



Get it away
from your
face and
keyboard

16

Your voice matters more than your face. You can switch off video if you have to, but if you lose sound, your class is over. You can use a headset with a microphone, but this can become uncomfortable after a couple of hours (particularly if you wear glasses). A Yeti or Samson Meteor microphone is more comfortable, but if you use one, it's worth \$30 to get a stand (because if it's on your desk, all your audience will hear is your typing).

Share work in real time



← HackMD or Google Doc

17

<https://unsplash.com/photos/Wh9ZavMYelw>

Share work in real time. Have learners share their work in a Google Doc or HackMD during the lesson. (This is no more or no less intimidating than asking a music student to play something for the class.) When you do this, paste in a list of people's names so that they know where to type. Once they're comfortable with this (which will take a few hours), have them take shared notes in the same doc so that your copilot can see what they think you're saying.

Keep track



- Yellow: names

<https://unsplash.com/photos/gdL-UZfnD3I>

Keep track of who has and hasn't spoken. Again, a laissez faire class can quickly turn into a discussion among a couple of extroverts, so keep a tally of who's spoken how often and try to even it out. You don't need technology to do this: a scrap of paper and a pencil will work just fine.

Keep track



- Yellow: names
- Red/green: status

<https://unsplash.com/photos/gdL-UZfnD3I>

Keep track of who has and hasn't spoken. Again, a laissez faire class can quickly turn into a discussion among a couple of extroverts, so keep a tally of who's spoken how often and try to even it out. You don't need technology to do this: a scrap of paper and a pencil will work just fine.

Keep track



- Yellow: names
- Red/green: status
- Blue: breaks

<https://unsplash.com/photos/gdL-UZfnD3I>

Keep track of who has and hasn't spoken. Again, a laissez faire class can quickly turn into a discussion among a couple of extroverts, so keep a tally of who's spoken how often and try to even it out. You don't need technology to do this: a scrap of paper and a pencil will work just fine.

Exercise

Describe one tool or practice you have used online (or seen used) that isn't included in this lesson.



What is one tool or practice you have used or seen used online that isn't covered in this lesson?